Project Proposal Report

Distance Based Dynamic Flood Risk Personalization and Multilingual Community Alerting for Kelani River Communities

Project Component Proposal

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March 2026

DECLARATION

I declare that this proposal is my own work and does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution. It also does not contain any material previously published or written by another person except where proper acknowledgement is given in the text.

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The above candidate is conducting research for the undergraduate dissertation under my supervision.

ABSTRACT

This project introduces a Dynamic Flood Risk Rating (DFRS) system by combining data from the Department of Irrigation and modern technology. Here, river level data received every 10 minutes, IoT sensors that measure rainfall intensity, elevations in the area and weather forecasts are all analyzed together. This allows us to accurately calculate the flood risk of a house or place based on the nature of the terrain.

Using the data calculated in this way, maps are prepared that identify people in the area who need to be evacuated and give them priority. Information about a hazard is provided to the public and relevant officials in three languages - Sinhala, Tamil and English through a mobile application (App), a computer control panel, SMS and voice calls. This is a personal warning method that goes beyond general notification and is given according to the location of each person.

In creating this system, React Native technology is used for the mobile application, Python for data analysis and advanced software such as GeoPandas for map preparation. The main result of this is that accurate information is quickly disseminated to the last person at risk of flooding.

Keywords: flood risk scoring, geospatial personalization, multilingual alerting, Kelani River, disaster management

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, lecturers and all stakeholders who provide guidance, technical insight and domain knowledge for this research. I also acknowledge the contribution of government agencies, community participants and data providers whose support is essential for the successful completion of this project.

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LIST OFABBREVIATIONS

DMC - Disaster Management Center

DFRS - Dynamic Food Risk Score

DTM - Displacement Tracking Matrix

GIS -Geographic Information System

GPS Global Positioning System

IOM - International Organization For Migration

IOT - Internet Of Things

1. INTRODUCTION

1.1 Background and Context

Flooding is one of the most disruptive natural hazards affecting riverine settlements in Sri Lanka. Communities near the Kelani River are especially vulnerable because rising river levels, concentrated local rainfall and delayed communication can quickly escalate into a life threatening situation. Although warning information exists, much of it is still presented at a macro level, making it difficult for households to understand how danger applies to their exact location.

This proposal addresses a last mile warning problem: authoritative hydrological observations and community level rainfall data are not being translated into household specific evacuation intelligence. As a result, two households in the same broad area may receive the same warning even when their actual exposure differs due to distance from the river, elevation and local rainfall intensity.

The proposed component will ingest 10 minute river level observations for Kelani River stations such as Nagalagam Street, Hanwella and Kithulgala from the Irrigation Department public dashboard, combine them with rainfall information from an IoT network and estimate each registered household's current and forecasted exposure. The system will then issue multilingual alerts and evacuation priority guidance that are understandable and actionable.

1.2 Literature Review

Research on disaster informatics shows that the value of a flood warning system depends not only on forecast quality but also on message relevance, timing and accessibility at the community level.

Hydrological monitoring systems often provide station level readings and threshold based warnings, which are useful for institutions but insufficient for households needing street level or property level guidance.

Geographical flood risk studies mostly look at how close a house is to rivers, how high the land is, and where it has flooded before to figure out the danger for each home. In the same way, studies on emergency apps show that it's important to have different languages, smooth ways to send

warnings and messages sent through many paths like app alerts, texts and phone calls. This ensures that info is shared clearly with all kinds of people. Right now, most websites and tools focus on the whole area or help big organizations monitor things, rather than helping individuals directly.

Existing dashboards and services mainly focus on regional awareness or institutional monitoring. Their major limitation is the absence of dynamic, personalized scoring for each household tied to authoritative river observations and terrain aware exposure.

1.3 Research Gap

The key research gap is the lack of a Sri Lankan system that transforms real time authoritative river data and local rainfall intelligence into household specific flood risk scores and evacuation instructions. Existing systems mainly provide area level warnings and do not adequately reach the grassroots level with personalized risk information.

This proposal is novel because it combines authoritative 10 minute river measurements and official thresholds with dynamic hydrological behavior such as the rate of rise, forecasted water levels from a prediction engine, static geospatial vulnerability factors such as horizontal distance and elevation and multilingual alert delivery in Sinhala, Tamil and English.

Existing Situation	Gap	Proposed Contribution
Flood warnings are mostly area-based	No household level personalization	Household specific DFRS
Warnings do not fully reach grassroots communities	No strong last mile alerting	Mobile, SMS, and voice alerts
Existing systems use limited factors	No integrated hydrological + geospatial scoring	Risk model using river, rainfall, distance, and elevation
Language accessibility is limited	No automated trilingual warning system	Sinhala, Tamil, English alerts

Table 1Research Gap

2. OBJECTIVE

2.1 Main Objective

To design and implement a household level dynamic flood risk personalization and multilingual community alerting component for Kelani River communities using real time river observations, rainfall intelligence and geospatial vulnerability factors.

2.2 Specific Objectives

Objective No.	Specific Objective
SO 1	To implement an automated client that retrieves 10 minute river level data for selected
SO 2	To build a geospatial household database with GPS coordinates and derive terrain aware
SO 3	To develop a Dynamic Flood Risk Score (DFRS) algorithm using current water level,
SO 4	To validate the DFRS against historical flood inundation extents and generate evacuation priority tiers for responders.
SO 5	To develop a multilingual alerting system and React Native mobile application with personalized push, SMS and voice based guidance for at risk households.
SO 6	To provide web based dashboards and a public portal that support institutional

Table 2 Specific Objectives

3. METHODOLOGY

This research will follow a Design Science and applied systems development approach. The study will produce and evaluate a functional artifact: a household level flood risk scoring and alerting component integrated with external flood prediction and monitoring sources.

Technical Methods and Algorithms

- Real Time Data: River levels are updated every 10 minutes using data from public sensors.
- Location Mapping: Every registered house is accurately located using GPS coordinates.
- Spatial Analysis: Tools like QGIS and GeoPandas measure the distance between houses and the river while analyzing the land's shape.
- Height Estimation: Digital maps (DEM) are used to calculate how high each house sits above the river.
- Risk Scoring: The DFRS formula determines flood danger by looking at water levels, how fast they rise, and weather forecasts.
- Data Validation: Results are compared with historical flood records from the DMC and IOM to ensure the system is accurate.

Tools and Platforms

Component	Proposed Tool/Platform	Justification
Mobile application	React Native	Cross-platform delivery of personalized alerts and household interaction

Spatial analytics	QGIS / GeoPandas	Reliable geospatial analysis for distance and elevation-related processing
Backend services	Python-based APIs and schedulers	Suitable for data ingestion, analytics and alert generation
Database	Geospatial-enabled relational database	Required for household coordinates, risk states and history
Dashboard / portal	Web application framework	Needed for institutional visualization and public access

Table 3 Tools and platforms

Data Collection and Ethical Considerations

The component will use river monitoring observations from the public Irrigation Department dashboard, rainfall information from the associated IoT network, registered household coordinates collected with consent, and historical inundation layers for validation. Household location and contact details will be handled as sensitive data. Access control, data minimization, encryption and consent-based registration will be applied.

Validation Strategy and Performance Metrics

Validation will compare model generated risk outputs with historical inundation extents and observed flood impact patterns . The success of the system depends on the accurate identification of flood risks, the speed at which data is updated and the successful delivery of warning messages to the public. It is also important that the system is easy to use by anyone.

4. HIGH LEVEL SYSTEM ARCHITECTURE

The proposed solution consists of six main layers: data acquisition, geospatial data storage, analytics and scoring, alert generation, institutional dashboards and public information access.

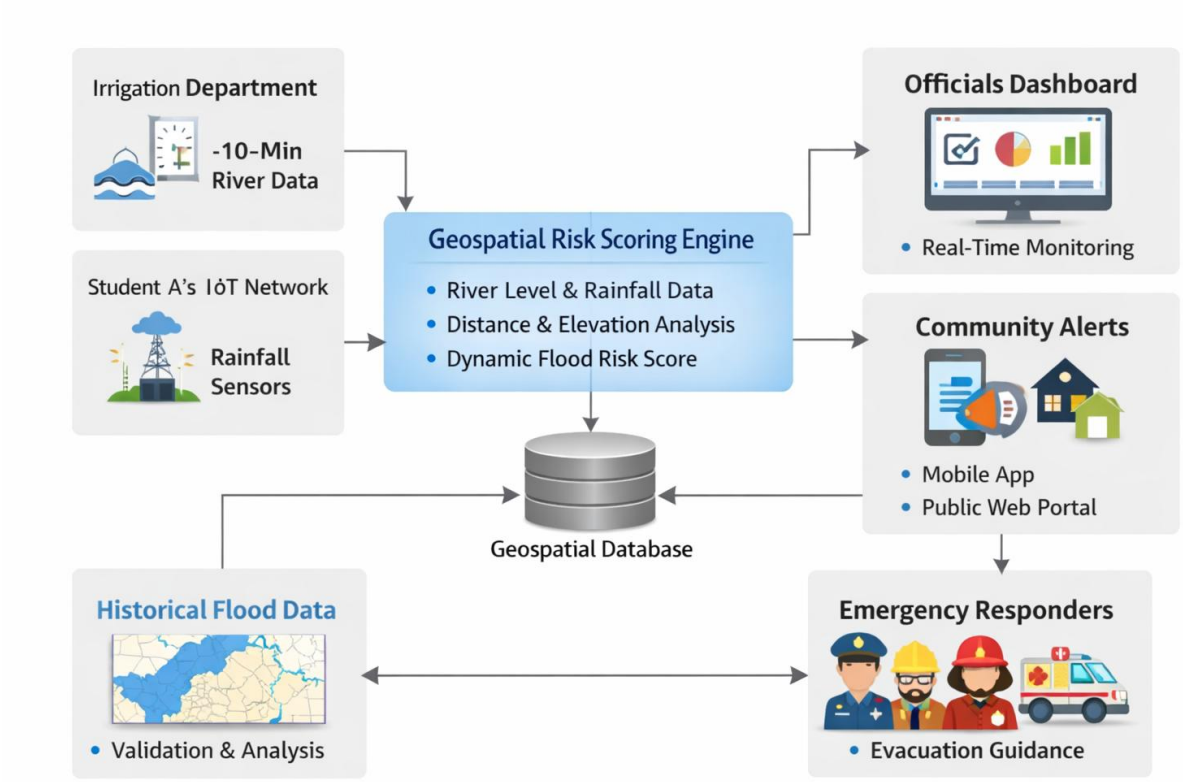


Figure 1 High Level System Architecture

Data flows from monitoring sources into the backend, where it is cleaned, synchronized and joined with household geospatial features. The DFRS engine calculates household specific risk scores and assigns evacuation priority tiers. Risk outputs are then routed to both communication channels and dashboard interfaces.

5. USER REQUIREMENTS

Stakeholder Identification

Primary stakeholders include at risk households in Kelani River communities, emergency responders, DMC officials, Irrigation Department officials and system administrators.

Functional Requirements

ID	Requirement
FR1	The system shall fetch 10 minute river level data from selected Kelani River stations automatically.
FR2	The system shall store registered household locations with GPS coordinates and profile details.
FR3	The system shall calculate household distance from river and elevation above river level.
FR4	The system shall compute DFRS values using real-time, forecasted and geospatial inputs.
FR5	The system shall assign dynamic evacuation priority tiers to households.
FR6	The system shall send personalized alerts in Sinhala, Tamil and English.
FR7	The mobile app shall support push notifications and household facing risk information.
FR8	The system shall support SMS and automated voice calls for high risk households.
FR9	The dashboard shall display sensor status, heatmaps, inundation zones and allocation priorities.
FR10	The public portal shall show colour coded flood risk by area.

Table 4 User Requirements

Non-Functional Requirements

Category	Requirement
Performance	Risk updates and alert generation should occur with minimal latency after new input data arrives.
Reliability	The system should tolerate temporary source failures and preserve recent valid data.
Usability	Interfaces should be simple enough for both officials and community users.
Security	Sensitive household data must be protected through authentication, authorization and encryption.
Scalability	The design should support expansion to additional river basins and larger user populations.
Accessibility	Alerts must be understandable across Sinhala, Tamil and English language users.

Table 5 Non Function Requrements

6. COMMERCIALIZATION PLAN

The main customers are disaster management agencies, local governments and regional authorities that require flood warnings and evacuation assistance. In addition, humanitarian relief organizations and private disaster relief projects are also expected to benefit from the services of this system. The value proposition is the conversion of raw flood observations into personalized, multilingual, location aware warning intelligence.

Revenue can be generated through software licensing, annual maintenance services and message issuance fees. Revenue can also be generated by modifying the system and expanding it to cover other disasters like landslides.The uniqueness of this approach is the provision of information at the household level, the issuance of messages in several languages, and the use of official

government data. The DFRS formula that calculates the risk and the functionality of the system are considered intellectual property here.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Purpose	Justification
Cloud hosting and storage	6000	Backend, database, portal, dashboard hosting	Required for continuous operation and testing
SMS / voice alert credits	4000	Pilot alert delivery	Needed to test real world communication workflows

Mapping / terrain data preparation	2500	Spatial preprocessing and validation	Supports distance and elevation based risk analysis
Mobile and web testing	2000	Usability and device/browser testing	Ensures stable experience across target users
Field data collection support	3000	Household registration and validation	Needed for geospatial household database creation
Contingency	2500	Unexpected technical or operational needs	Provides implementation flexibility

Table 6 Budget and justification

Estimated Total Budget: LKR 20 000

8. WORK BREAKDOWN STRUCTURE (WBS)

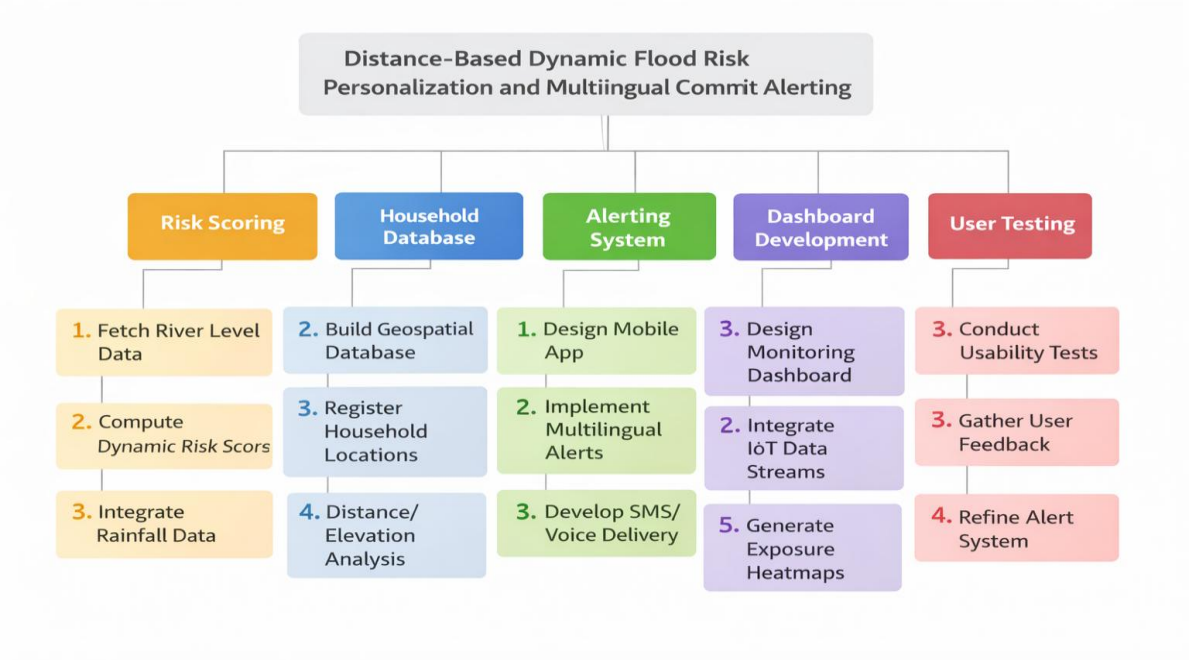


Figure 2 Work Breakdown Chart

9. GANTT CHART

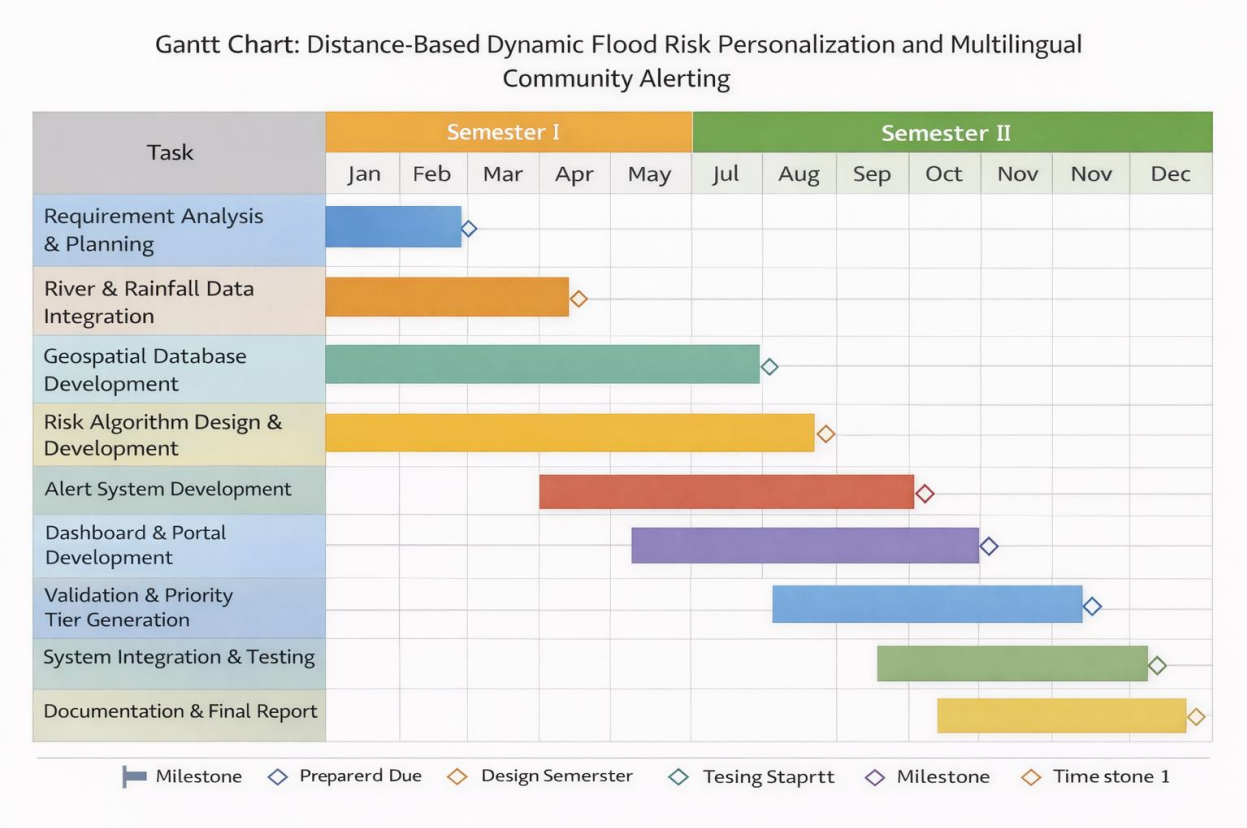


Figure 3 Gantt Chart

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